Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent Student code: _____ Calculator: no

Learning goal: I can tell whether two expressions are equivalent. (circle your level after scoring)

4	I can justify equivalence using properties of operations, including factoring.
3	I can decide whether two linear expressions are equivalent and rewrite one as the other.
2	I can use the distributive property on a simple expression.
1	I need help to start this benchmark.

1. Select **all** the expressions that are equivalent to $\left(\frac{5}{4}\right)(12x + 20)$.

Select **all** the correct answers.

A. ☐ $\left(\frac{5}{4}\right)(12x) + \left(\frac{5}{4}\right)(20)$

B. ☐ $\left(\frac{53}{4}\right)x + \frac{85}{4}$

C. ☐ $15x + 25$

D. ☐ $15x + 20$

E. ☐ $40x$

2. Match each expression to an equivalent expression.

Check one box in each row.

	$5x - 4$	$7x + 4$	$20x + 15$
$2.5(8x + 6)$	☐	☐	☐
$\left(\frac{1}{2}\right)(10x - 8)$	☐	☐	☐
$(2.5x + 6) + (4.5x - 2)$	☐	☐	☐

Turn over →

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent Student code: _____

3. Which expression is equivalent to $1.5(6x - 2)$?

A. $9x - 2$

B. $9x - 3$

C. $6x$

D. $7.5x - 0.5$

4. Are the two expressions equivalent?

$2.5(8x - 2) + 4x + 8$ and $24x + 3.5$

A. Yes, because both of them simplify to $24x + 3.5$.

B. No, because $2.5(8x - 2)$ is $20x - 2$, giving $24x + 6$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $24x + 3$, not $24x + 3.5$.

5. Two expressions are shown.

$-\left(\frac{3}{4}\right)(4x - 4) + \left(\frac{3}{2}\right)x$ and $\left(\frac{11}{2}\right)x - \left(\frac{3}{4}\right)(8x - 4)$

Are the two expressions equivalent? Write each expression using the fewest possible terms, and use those forms to explain your answer.

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-AR.1.2 · MA.7.AR.1.2 · Determine if two expressions are equivalent

Unit 9: Algebraic Expressions. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **A.** $\left(\frac{5}{4} = 1\frac{1}{4}\right)(12x) + \left(\frac{5}{4} = 1\frac{1}{4}\right)(20)$**; C.** $15x + 25$

Distribute $\frac{5}{4}$ to both terms: $\frac{5}{4} \times 12x = 15x$ and $\frac{5}{4} \times 20 = 25$.
So $\left(\frac{5}{4}\right)(12x + 20) = 15x + 25$, which is also written $\left(\frac{5}{4}\right)(12x) + \left(\frac{5}{4}\right)(20)$.
Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

2. Regular $2.5(8x + 6) \rightarrow 20x + 15$**;** $\left(\frac{1}{2}\right)(10x - 8) \rightarrow 5x - 4$**;** $(2.5x + 6) + (4.5x - 2) \rightarrow 7x + 4$

Use the properties of operations to write each expression with the fewest possible terms.
$\left(\frac{1}{2}\right)(10x - 8) = 5x - 4$
$(2.5x + 6) + (4.5x - 2) = 7x + 4$
$2.5(8x + 6) = 20x + 15$
Scoring: 1 point only if every match is correct; anything else earns 0.

3. Regular **B.** $9x - 3$

Distribute 1.5 to both terms inside the parentheses.
$1.5 \times 6x = 9x$ and $1.5 \times (-2) = -3$, so the result is $9x - 3$.

4. Regular **D. No, because the first simplifies to $24x + 3$, not $24x + 3.5$.**

Distribute and combine: $2.5(8x - 2) = 20x - 5$, then + $4x + 8$ gives $24x + 3$.
The second expression is $24x + 3.5$. The constants differ (3 and 3.5), so they are not equivalent.
The 2.5 multiplies both terms in the parentheses.

5. Rigor **No; the first expression simplifies to $-\left(\frac{3}{2} = 1\frac{1}{2}\right)x + 3$ and the second to $-\left(\frac{1}{2}\right)x + 3$. Score: 1 point for the correct yes or no with both simplified forms; yes or no alone earns 0.**

First expression: distribute $-\frac{3}{4}$: $-\frac{3}{4} \times 4x = -3x$ and $-\frac{3}{4} \times (-4) = 3$. Combine the x-terms: $-3x + \left(\frac{3}{2}\right)x = -\left(\frac{3}{2}\right)x$. So the first expression is $-\left(\frac{3}{2}\right)x + 3$.
Second expression: distribute $-\frac{3}{4}$: $-\frac{3}{4} \times 8x = -6x$ and $-\frac{3}{4} \times (-4) = 3$. Combine the x-terms: $\left(\frac{11}{2}\right)x - 6x = -\left(\frac{1}{2}\right)x$. So the second expression is $-\left(\frac{1}{2}\right)x + 3$.
The constants match but the x-terms differ, so the expressions are not equivalent.

Re-check strip (3 items, C-AR.1.2-R)

1. Regular **D. No, because the first simplifies to $7x - 1$, not $7x - 2$.**

Distribute and combine: $1.5(4x - 6) = 6x - 9$, then + $x + 8$ gives $7x - 1$.
The second expression is $7x - 2$. The constants differ $(-1 \text{ and } -2)$, so they are not equivalent.
The 1.5 multiplies both terms in the parentheses.

2. Regular **A.** $1.5(4x + 5)$

Both terms of $6x + 7.5$ are 1.5 times a whole number: $6 = 1.5 \times 4$ and $7.5 = 1.5 \times 5$.
So $6x + 7.5 = 1.5(4x + 5)$. Check by distributing: $1.5 \times 4x = 6x$ and $1.5 \times 5 = 7.5$.

3. Regular **A.** $\left(\frac{3}{2} = 1\frac{1}{2}\right)(6x) + \left(\frac{3}{2} = 1\frac{1}{2}\right)(2)$**; E.** $9x + 3$

Distribute $\frac{3}{2}$ to both terms: $\frac{3}{2} \times 6x = 9x$ and $\frac{3}{2} \times 2 = 3$.
So $\left(\frac{3}{2}\right)(6x + 2) = 9x + 3$, which is also written $\left(\frac{3}{2}\right)(6x) + \left(\frac{3}{2}\right)(2)$.
Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Comparison strip (2 items, C-AR.1.2-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B.** $\left(\frac{2}{5}\right)(20x + 15)$

Both terms of $8x + 6$ are $\frac{2}{5}$ times a whole number: $8 = \frac{2}{5} \times 20$ and $6 = \frac{2}{5} \times 15$.
So $8x + 6 = \left(\frac{2}{5}\right)(20x + 15)$. Check by distributing: $\frac{2}{5} \times 20x = 8x$ and $\frac{2}{5} \times 15 = 6$.
$\left(\frac{5}{2}\right)(20x + 15)$ uses the reciprocal factor and does not give $8x + 6$.

2. Regular $1.5(2x + 2) \rightarrow 3x + 3$**;** $\left(\frac{1}{4}\right)(16x - 8) \rightarrow 4x - 2$**;** $(2.5x + 7) + (1.5x - 5) \rightarrow 4x + 2$

Use the properties of operations to write each expression with the fewest possible terms.
$\left(\frac{1}{4}\right)(16x - 8) = 4x - 2$
$(2.5x + 7) + (1.5x - 5) = 4x + 2$
$1.5(2x + 2) = 3x + 3$
Scoring: 1 point only if every match is correct; anything else earns 0.

Reteach plan — C-AR.1.2

MA.7.AR.1.2 · Determine if two expressions are equivalent

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-AR.1.2-R

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent Student code: _____ Calculator: no

1. Are the two expressions equivalent?

$1.5(4x - 6) + x + 8$ and $7x - 2$

A. No, because $1.5(4x - 6)$ is $6x - 6$, giving $7x + 2$.

B. Yes, because both of them simplify to $7x - 2$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $7x - 1$, not $7x - 2$.

2. Which expression is equivalent to $6x + 7.5$?

A. $1.5(4x + 5)$

B. $6(x + 5)$

C. $1.5(6x + 5)$

D. $1.5(4x + 7.5)$

3. Select **all** the expressions that are equivalent to $\left(\frac{3}{2}\right)(6x + 2)$.

Select **all** the correct answers.

A. ☐ $\left(\frac{3}{2}\right)(6x) + \left(\frac{3}{2}\right)(2)$

B. ☐ $\left(\frac{15}{2}\right)x + \frac{7}{2}$

C. ☐ $12x$

D. ☐ $9x + 2$

E. ☐ $9x + 3$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-AR.1.2-R

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent Student code: _____ Calculator: no

1. Are the two expressions equivalent?

$1.5(4x - 6) + x + 8$ and $7x - 2$

A. No, because $1.5(4x - 6)$ is $6x - 6$, giving $7x + 2$.

B. Yes, because both of them simplify to $7x - 2$.

C. Yes, because both have an x-term and a constant term.

D. No, because the first simplifies to $7x - 1$, not $7x - 2$.

2. Which expression is equivalent to $6x + 7.5$?

A. $1.5(4x + 5)$

B. $6(x + 5)$

C. $1.5(6x + 5)$

D. $1.5(4x + 7.5)$

3. Select **all** the expressions that are equivalent to $\left(\frac{3}{2}\right)(6x + 2)$.

Select **all** the correct answers.

A. ☐ $\left(\frac{3}{2}\right)(6x) + \left(\frac{3}{2}\right)(2)$

B. ☐ $\left(\frac{15}{2}\right)x + \frac{7}{2}$

C. ☐ $12x$

D. ☐ $9x + 2$

E. ☐ $9x + 3$

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-AR.1.2-X

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent

Student code: _____ Calculator: no

1. Which expression is equivalent to $8x + 6$?

A. $\left(\frac{2}{5}\right)(20x + 6)$

B. $\left(\frac{2}{5}\right)(20x + 15)$

C. $\left(\frac{5}{2}\right)(20x + 15)$

D. $\left(\frac{2}{5}\right)(8x + 15)$

2. Match each expression to an equivalent expression.

Check one box in each row.

	$4x - 2$	$4x + 2$	$3x + 3$
$1.5(2x + 2)$	☐	☐	☐
$\left(\frac{1}{4}\right)(16x - 8)$	☐	☐	☐
$(2.5x + 7) + (1.5x - 5)$	☐	☐	☐

Score: ____ / 2 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-AR.1.2-X

Benchmark: **MA.7.AR.1.2** · Determine if two expressions are equivalent

Student code: _____ Calculator: no

1. Which expression is equivalent to $8x + 6$?

A. $\left(\frac{2}{5}\right)(20x + 6)$

B. $\left(\frac{2}{5}\right)(20x + 15)$

C. $\left(\frac{5}{2}\right)(20x + 15)$

D. $\left(\frac{2}{5}\right)(8x + 15)$

2. Match each expression to an equivalent expression.

Check one box in each row.

	$4x - 2$	$4x + 2$	$3x + 3$
$1.5(2x + 2)$	☐	☐	☐
$\left(\frac{1}{4}\right)(16x - 8)$	☐	☐	☐
$(2.5x + 7) + (1.5x - 5)$	☐	☐	☐

Score: ____ / 2 → **copy to your progress strip**